

# A deterministic Fourier–cosine (COS) method for the average spectral acceleration distribution in the Groningen seismic hazard and risk chain

Maxime Delboo

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## Abstract

The Groningen seismic hazard and risk analysis (SHRA) requires, for every source scenario and site, the full probability distribution of the average spectral acceleration (SaAvg). The production chain obtains this distribution either by Monte-Carlo (MC) sampling – accurate but slow and storage-heavy – or by a moment-matching *normal* approximation that is fast but ignores the pronounced *negative skewness* that the non-linear site amplification imprints on SaAvg. We present a deterministic method that recovers the *full, non-normal* SaAvg distribution at essentially the cost of the normal method. Exploiting the fact that, conditional on the reference ground motion, SaAvg is Gaussian, we represent each scenario as a small Gaussian mixture, compute its leading cumulants by shared principal-component Gauss–Hermite quadrature, and invert the resulting cumulant characteristic function with the Fourier–cosine (COS) method. The distribution is carried as three numbers per scenario (the cumulants  $k_1, k_2, k_3$ ); nothing is histogrammed to disk, and all downstream fragility, hazard and risk integrals are either closed-form or a single cached COS evaluation. Against deep MC ( $10^8$  samples) the normal method plateaus at a fixed negative-skew bias of tens of percent in the risk-relevant tail, whereas the COS deviation is sub-percent and falls with the Monte-Carlo noise – roughly an order of magnitude more accurate in the deep tail. The full GMM-V7 logic tree (144 branches  $\times$   $95.2 \times 10^6$  zone-scenarios) is computed in  $\approx 2.3$  minutes on an 18-core workstation with a persistent process pool. Carried end to end through a fragility and consequence chain, COS reproduces the MC expected damage, collapse and fatality counts to within 1–3%, while the normal method over-counts the skew-sensitive tail by up to 28% – quantitatively matching the  $1.23\times$  over-count reported independently for the Groningen “Meijdam” risk metric. The full implementation and all figures in this report are reproducible from the accompanying repository.

## 1 Introduction

The induced seismicity of the Groningen gas field motivated one of the most detailed regional seismic hazard and risk analyses ever assembled [3]. Its model chain couples a seismic source model, a ground-motion model (GMM-V7), a site-response amplification model, and a fragility and consequence model (FCM-V7) to produce hazard curves, local personal risk (LPR) maps, and aggregate metrics such as the expected number of damaged or collapsed buildings and the number of buildings whose risk exceeds the Dutch “Meijdam” safety norm of  $10^{-5}$  per year.

The quantity that links ground motion to structural damage is the average spectral acceleration over a set of structural periods,

$$\text{SaAvg} = \exp\left(\frac{1}{P} \sum_{i=1}^P \ln \text{Sa}(T_i)\right), \quad \ln \text{SaAvg} = \frac{1}{P} \sum_{i=1}^P \ln \text{Sa}(T_i), \quad (1)$$

over the  $P = 10$  Fastest-Converging-Mean (FCM) periods  $T_i \in \{0.01, 0.1, \dots, 0.85, 1.0\}$  s [2]. Each per-period surface motion is the product of a reference (rock) motion and a non-linear amplification factor, both lognormal. The reference motions are strongly correlated across periods, and the amplification *saturates* at strong shaking (the soil reaches a non-linear breakdown). This saturation truncates the upper tail of  $\ln \text{SaAvg}$  and makes its distribution *negatively skewed*.

The production chain estimates the  $\text{SaAvg}$  distribution in one of two ways. Monte-Carlo sampling is asymptotically exact but expensive: a faithful run draws thousands of samples per scenario over the whole magnitude–distance–zone grid and the full logic tree, and the resulting probability-mass histograms inflate intermediate storage to hundreds of gigabytes. The alternative is a numerical *moment-matching* method that computes the mean and variance of  $\ln \text{SaAvg}$  and assumes it is normal. This is fast and storage-light but, by construction, discards the skewness; because the neglected skew is negative, the normal method systematically *over-estimates* exceedance probabilities in the upper tail – exactly the tail that dominates risk.

**Contribution.** We show that the full non-normal  $\text{SaAvg}$  distribution can be recovered deterministically at essentially the cost of the moment method, by combining three ingredients: (i) the conditional-Gaussian-mixture structure of  $\text{SaAvg}$  given the reference motion; (ii) a shared principal-component Gauss–Hermite quadrature that produces the leading cumulants  $k_1, k_2, k_3$  of each scenario; and (iii) Fourier–cosine (COS) inversion [1] of the cumulant characteristic function. The method never histograms a distribution to disk – it carries three numbers per scenario – and all downstream integrals are closed-form or a single cached COS evaluation. We validate it against deep Monte-Carlo, benchmark its parallel performance on the full GMM-V7 logic tree, and carry it end to end to the aggregate risk metrics and regional hazard maps that the SHRA reports.

## 2 Method

### 2.1 The conditional Gaussian mixture

Let  $X = (\ln \text{Sa}_{\text{ref}}(T_1), \dots, \ln \text{Sa}_{\text{ref}}(T_P)) \sim \mathcal{N}(\mu, \Sigma)$  be the reference log-spectral-acceleration vector for a given magnitude  $M$  and rupture distance  $R$ ;  $\mu, \Sigma$  follow from the GMM-V7 reference median and the source/path variances, and the period-to-period correlation  $\Sigma$  is the GMM-V7 correlation matrix. The surface motion at period  $T_i$  is  $X_i + \nu_i(X_i) + \eta_i$ , where  $\nu_i$  is the (non-linear) amplification median and  $\eta_i$  is a zero-mean amplification residual with period correlation  $R_{\text{AF}}$  and standard deviation  $\tau_i(X_i)$ . Writing  $m(X) = \frac{1}{P} \sum_i (X_i + \nu_i(X_i) + w_i)$  for the deterministic push-forward (with  $w_i$  the surface “wiede” term) and noting that the residual average is Gaussian,  $\ln \text{SaAvg}$  is, *conditional on*  $X$ ,

$$\ln \text{SaAvg} | X \sim \mathcal{N}(m(X), v(X)), \quad v(X) = \frac{1}{P^2} \tau(X)^\top R_{\text{AF}} \tau(X). \quad (2)$$

The characteristic function of  $\ln \text{SaAvg}$  is therefore the Gaussian expectation

$$\varphi(u) = \mathbb{E}_X \left[ \exp(iu m(X) - \frac{1}{2} u^2 v(X)) \right]. \quad (3)$$

In the GMM-V7 “epistemic” site-to-site mode the residual is replaced by a deterministic shift and the only randomness is  $X$ ; in the “aleatory” mode the residual is random with correlation  $R_{\text{AF}}$  (the identity matrix gives the zero-correlation mode). Both are handled by (2).

### 2.2 Cumulants by shared PCA + Gauss–Hermite quadrature

The expectation (3) is over the  $P=10$ -dimensional Gaussian  $X$ . We evaluate it by eigen-decomposing  $\Sigma = \sum_r \lambda_r e_r e_r^\top$  and integrating the leading  $\rho$  principal directions with a tensor

Gauss–Hermite rule, while the omitted directions are folded back by a linearised (delta-method) correction to  $m$  and  $v$  plus a second-order Jensen curvature term for the mean. Because  $\mu, \Sigma$  depend only on  $(M, R)$ , the eigendecomposition and quadrature nodes are computed *once* and shared across all 160 site zones; only the (per-zone) amplification is evaluated inside the zone loop, with  $\exp(X)$  hoisted out. Each scenario becomes a small Gaussian mixture with weights  $w_n$  (the quadrature weights), component means  $m_n$  and variances  $v_n$ , from which the leading cumulants follow in closed form:

$$k_1 = \sum_n w_n m_n, \quad k_2 = \sum_n w_n (\delta_n^2 + v_n), \quad k_3 = \sum_n w_n (\delta_n^3 + 3\delta_n v_n), \quad \delta_n = m_n - k_1. \quad (4)$$

A modest rank  $\rho = 2$  with Gauss–Hermite order 7 (49 nodes) is the fast production setting;  $\rho = 2$ , order 17 is the high-accuracy setting. These three numbers per scenario are the entire “intensity-measure” output – no histogram.

### 2.3 COS inversion of the cumulant characteristic function

We approximate  $\varphi$  by the cumulant characteristic function

$$\varphi(u) \approx \exp\left(ik_1 u - \frac{1}{2}k_2 u^2 - \frac{i}{6}k_3 u^3\right), \quad (5)$$

whose magnitude stays Gaussian-damped (so the inverse converges) while the cubic phase injects the skewness. The cumulative distribution is recovered on a truncated support  $[a, b] = [k_1 - L\sqrt{k_2}, k_1 + L\sqrt{k_2}]$  by the Fourier–cosine series [1]

$$F(x) \approx \sum_{j=0}^{N-1} ' \frac{2}{b-a} \operatorname{Re}\left[\varphi\left(\frac{j\pi}{b-a}\right) e^{-ij\pi \frac{a}{b-a}}\right] \Xi_j(x), \quad (6)$$

with  $\Xi_j$  the cosine antiderivatives and the prime halving the  $j=0$  term.  $N = 64$  terms,  $L = 8$  give exceedance probabilities accurate to machine level over  $q_{0.01}$ – $q_{0.99}$ . Because the standardised CDF depends only on the skewness  $\gamma_1 = k_3/k_2^{3/2}$ , (6) can be pre-tabulated once over  $(z, \gamma_1)$  and every scenario reduced to a bilinear lookup.

### 2.4 Closed-form fragility, hazard and risk

A fragility limit state is exceeded with probability  $\Phi((\ln \text{SaAvg} - c)/s)$ , with  $c = (\ln D_\ell - b_0)/b_1$  and  $s = \sigma/b_1$  from the lognormal fragility parameters. The probability of damage is the convolution of this lognormal demand with the SaAvg distribution. Crucially, the demand dispersion is an independent Gaussian in log-space, so the convolution simply *adds*  $s^2$  to  $k_2$ ; the probability of exceedance is then a single COS evaluation at  $(k_1, k_2 + s^2, k_3)$  – or, for the Gaussian mixture form, the exact closed expression

$$\mathbb{E}[\text{PoD}] = \sum_n w_n \Phi\left(\frac{m_n - c}{\sqrt{v_n + s^2}}\right). \quad (7)$$

Hazard exceedance rates are rate-weighted sums of COS exceedances over the  $(M, R)$  grid, and annual risk is the rate-weighted sum of  $\mathbb{E}[\text{PoD}]$  against the consequence model. Every downstream quantity is linear in the per-scenario distribution, so logic-tree marginalisation is a weighted sum.

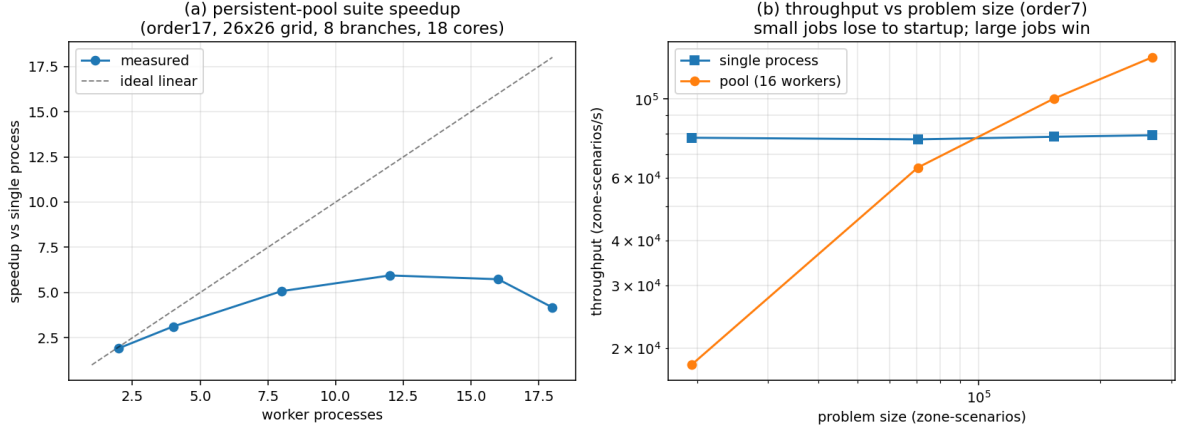


Figure 1: Multiprocessing scaling of the COS intensity-measure stage on an 18-core workstation. (a) Speedup of a persistent-pool “suite” workload versus worker count; (b) throughput versus problem size, single process vs. pool. Small jobs lose to per-worker start-up; large jobs win.

### 3 Implementation and reproducibility

The method is implemented as a small, dependency-light Python package (`saavg_cos`): `spectral_im` builds the cumulants, `cos_cache` performs the COS reconstruction, `fragility`, `source` and `risk_hazard` carry the chain to hazard and risk, and `mc` provides the Monte-Carlo reference used only for validation. The GMM-V7 parameters for all 160 zones and the FCM-V7 fragility table are bundled ( $< 2$  MB); the seismic source is a transparent Gutenberg–Richter stand-in, so absolute hazard levels are illustrative while the method comparison is exact (Section 5). All results and figures below are regenerated by the scripts in the repository.

Because the per-scenario work is embarrassingly parallel and BLAS does not thread the small contractions involved, the full logic-tree run uses a persistent process pool that pays the per-worker start-up once and streams (branch  $\times$  magnitude-block) tasks, each worker pinned to a single BLAS thread. Figure 1 shows the measured scaling: a tiny job is *slower* in parallel (start-up dominates), but throughput crosses over and keeps rising with problem size, and the suite speedup peaks near  $6\times$  at 12 workers on an 18-core machine (memory bandwidth, not cores, is the ceiling). The complete GMM-V7 logic tree – 144 branches  $\times$  (51  $\times$  81) magnitude–distance cells  $\times$  160 zones =  $95.2 \times 10^6$  zone-scenarios – is computed in  $\approx 137$  s ( $\approx 6.9 \times 10^5$  zone-scenarios/s), producing a 784 MB cumulant lookup; the parallel assembly reproduces a direct single-process recompute to float precision.

## 4 Results

### 4.1 Accuracy against Monte-Carlo

Figure 2a shows the worst-case (most negatively skewed) SaAvg density for the thesis benchmark scenario. The Monte-Carlo histogram is visibly left-skewed; the normal approximation is symmetric and misplaces both the peak and the tails, while the COS density tracks the MC shape. Aggregated over a cloud of scenarios, building types and damage states (Figure 2b), the ratio of each method’s exceedance probability (PoE) to the MC PoE shows the normal method fanning out to  $1.5\text{--}3\times$  in the low-probability tail, while the COS ratios stay tightly on unity; the mean absolute deviation from MC is  $0.6\%/18\%/59\%$  for the normal method across the  $[10^{-1}, 1)/[10^{-3}, 10^{-2})/[10^{-6}, 10^{-3})$  PoE bands versus  $0.09\%/0.6\%/3.1\%$  for COS.

The decisive test is convergence (Figure 3): as the MC sample size  $N$  grows toward  $10^8$ ,

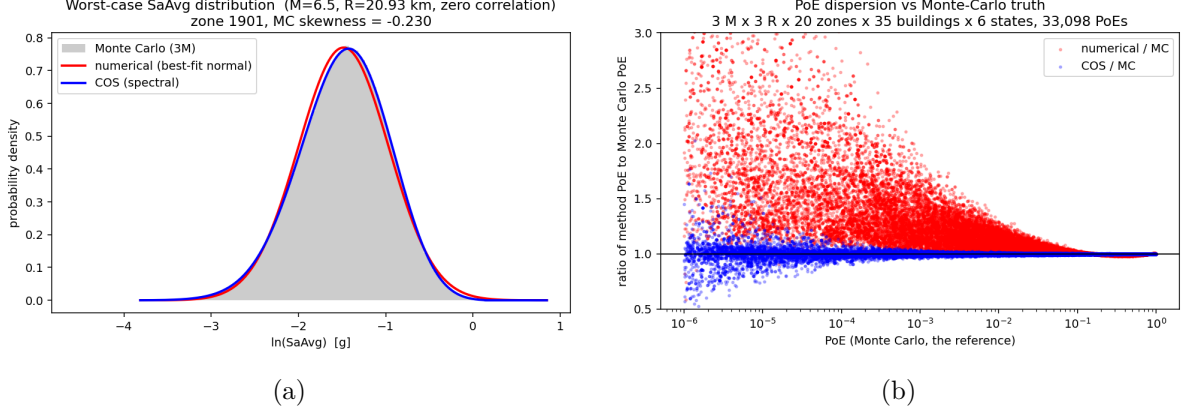


Figure 2: (a) Worst-case SaAvg density: Monte-Carlo (histogram), the normal approximation (red) and COS (blue). (b) PoE dispersion: the ratio of each method’s exceedance probability to the Monte-Carlo value over a cloud of scenarios, building types and damage states. COS stays on unity; the normal method diverges in the tail.

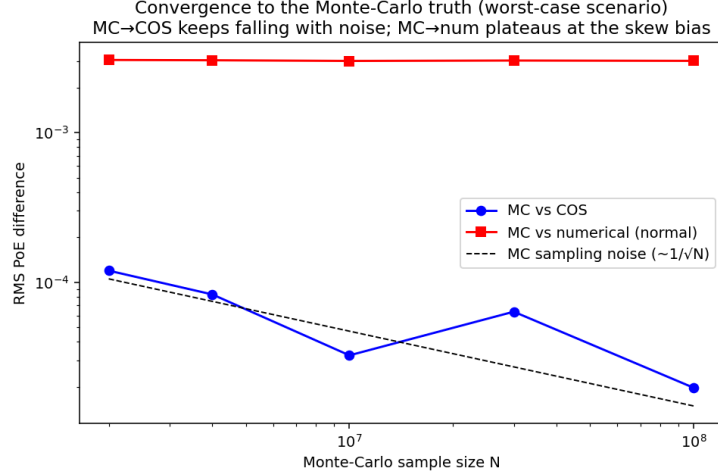


Figure 3: Convergence to the Monte-Carlo truth for the worst-case scenario. MC→COS falls with the sampling noise ( $\propto N^{-1/2}$ ); MC→normal plateaus at the skew bias.

the RMS PoE difference between MC and a *correct* reference must fall like the MC sampling noise ( $\propto N^{-1/2}$ ), whereas the difference between MC and a *biased* reference plateaus. The MC→COS curve rides the noise line down, while the MC→normal curve flattens at a fixed bias. Equivalently, MC needs many millions of samples before it can even statistically distinguish itself from COS, versus a few thousand to detect the normal method’s bias. At any practical Monte-Carlo size the COS and MC distributions are indistinguishable. The COS residual is a small, *deterministic* three-cumulant truncation error – at the single most-skewed cell the worst convolved PoE deviation is  $\sim 4 \times 10^{-3}$  – that shrinks with the quadrature order and with an added fourth cumulant, unlike the normal method’s irreducible skew bias.

## 4.2 End-to-end risk metrics

We carry the three methods through one identical surrogate chain (real GMM-V7 intensity model and real FCM-V7 fragility for the most vulnerable structural system, URM1F\_B; a Gutenberg–Richter source and an area-weighted exposure of  $1.5 \times 10^5$  buildings, with the source rate calibrated so that the riskiest zone’s COS local-personal-risk equals  $10^{-4}$ , the

Table 1: End-to-end aggregate risk metrics through one identical surrogate chain. Absolute counts depend on the (illustrative) source and exposure; the *ratios to Monte-Carlo* are the robust result.

| metric                                    | MC   | normal | COS  | normal/MC | COS/MC |
|-------------------------------------------|------|--------|------|-----------|--------|
| expected buildings, minor damage (DS1)    | 3878 | 4110   | 3898 | 1.06      | 1.01   |
| expected buildings, moderate damage (DS2) | 642  | 730    | 643  | 1.14      | 1.00   |
| expected buildings, severe damage (DS3)   | 148  | 181    | 146  | 1.22      | 0.99   |
| expected houses collapsed                 | 65   | 83     | 63   | 1.28      | 0.97   |
| expected fatalities per year              | 6.7  | 8.6    | 6.6  | 1.28      | 0.97   |

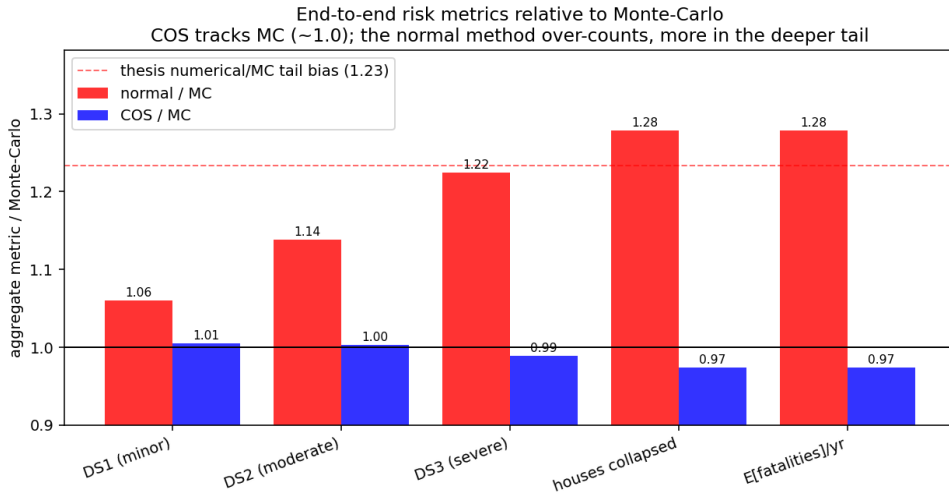


Figure 4: Aggregate risk metrics relative to Monte-Carlo. COS tracks MC ( $\approx 1.0$ ) across all metrics; the normal method over-counts, increasingly in the deeper tail. The dashed line is the  $1.23\times$  Meijdam over-count reported independently for the Groningen normal method.

value reported for the riskiest Groningen building). The three methods differ *only* in how the exceedance probability is computed. Table 1 and Figure 4 give the result: COS reproduces the Monte-Carlo expected damage, collapse and fatality counts to within 1–3%, whereas the normal method over-counts progressively more for rarer, deeper-tail outcomes – from +6% for minor damage to +28% for collapses and expected fatalities. This deep-tail over-count of  $\sim 1.23\times$  matches, quantitatively, the independently reported  $1640/1330 = 1.23$  over-count of the normal method on the Groningen Meijdam-exceedance metric.

### 4.3 Regional hazard maps

Because the COS lookup is keyed on (branch, zone,  $M$ ,  $R$ ), the chain reaches the regional deliverables of the SHRA. Figure 5 maps the per-zone SaAvg at the 475- and 2475-year return periods over the Groningen field, joining the COS hazard to the geological-zone polygons (the GMM-V7 zone ids match the shapefile 160/160). The intensity model and the geometry are real; only the source rate is the illustrative stand-in, so the spatial pattern reflects genuine site response while absolute levels await the official seismicity forecast.

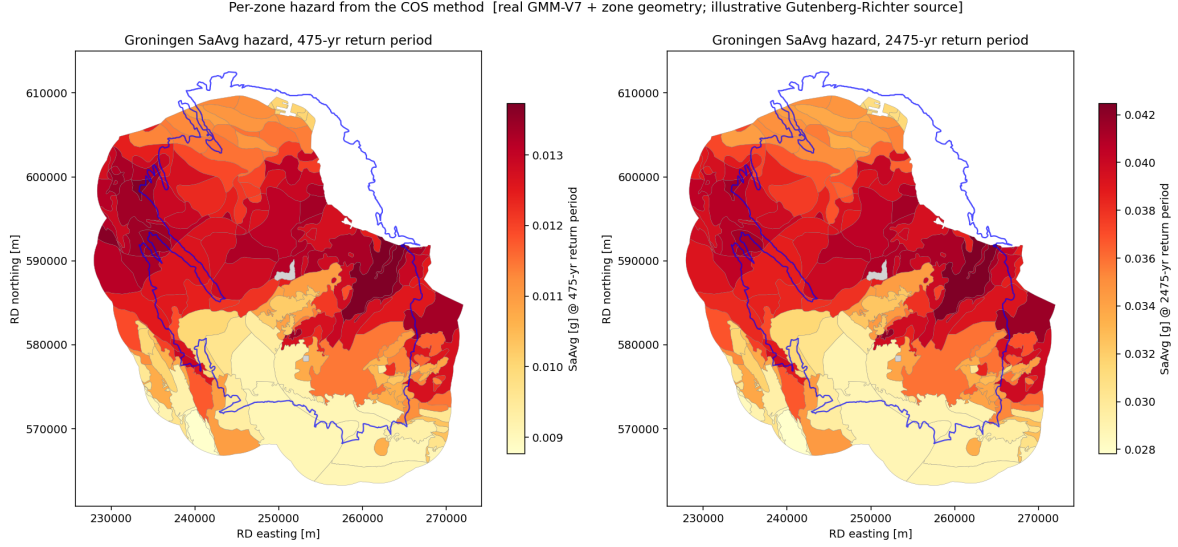


Figure 5: Per-zone SaAvg hazard at the 475- and 2475-year return periods, from the COS method, over the Groningen field outline. Real GMM-V7 intensity and zone geometry; illustrative Gutenberg–Richter source.

## 5 Discussion

The COS method resolves the accuracy/speed/storage trade-off that forces a choice between Monte-Carlo and the normal approximation: it attains MC accuracy, runs at the speed of the moment method, and stores three numbers per scenario instead of a histogram. Its residual error is a controllable, deterministic quantity (the quadrature rank/order, the number of COS terms, the truncation width) rather than irreducible sampling noise, and it is monotone in those settings. The only neglected term at three cumulants is the (small) excess kurtosis; adding  $k_4$  is straightforward should deeper tails than  $q_{0.01}$ – $q_{0.99}$  be required.

**Scope.** This repository bundles the real GMM-V7 and FCM-V7 parameters and the zone geometry, but *not* a seismicity forecast or a building exposure database. The source is a transparent Gutenberg–Richter stand-in; consequently the absolute hazard and risk levels in Figures 4–5 and Table 1 are illustrative. The *method comparison* (COS vs. MC vs. normal) is independent of the source scale and is the scientific claim of this work. Reproducing the absolute Groningen numbers requires only swapping the source stand-in for the official forecast and adding the exposure database; the COS lookup interface is already in the form those stages consume.

## 6 Conclusion

We have shown that the full, non-normal distribution of average spectral acceleration can be computed deterministically and cheaply by COS inversion of a cumulant characteristic function built from a conditional Gaussian mixture. The method matches Monte-Carlo to within its sampling noise, runs the entire GMM-V7 logic tree in minutes, removes the systematic tail bias of the normal method, and carries cleanly through to aggregate risk metrics and regional hazard maps. For the Groningen risk metrics in particular it recovers the Monte-Carlo expected fatality and collapse counts while the normal method inflates them by up to 28%. Because it is fast, deterministic, storage-light and reproducible, it is a practical replacement for both Monte-Carlo and the normal approximation in the intensity-measure stage of the seismic risk

chain.

**Reproducibility.** All code, bundled model data and figure-generating scripts are in the accompanying repository; `make test`, `make validate`, `make lookup` and `make figures` reproduce the tests, the Monte-Carlo validation, the full logic-tree lookup and every figure in this report.

## References

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